# Reduced basis methods for the resolution of parameter-dependent PDEs
# MS13

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**Goal: Reduce the computational cost of solving a parametric problem while preserving high accuracy**

https://grosjean1.github.io

1 Introduction

2 Reduced Basis Methods

3 Galerkin projection

4 POD + TP VF

# Introduction

- ◇ Understand the purpose of reduced basis methods and how they work
- ◇ Get a non-exhaustive overview of existing methods
- ◇ Implement different methods to gain better insight
- ◇ Understand the limitations and in some cases, how to overcome them

# Introduction

**Course objectives:** Become familiar with reduced basis methods and implement it (during TP sessions).

**6 lectures/TP sessions on Tuesday afternoons**

February 17: introduction + reduced basis approximation,
February 24: approximation space using POD (and VFM), TP1-TP2 (start),
March 10: POD + error estimation (with FEM), TP3-TP4 (continuation),
March 17: Greedy + Weak greedy, TP5 (continuation)
**Deadline for TP 1 report (25% of the grade): Monday, March 24, 11:59 PM**,
March 24: RB method for nonlinear PDEs, TP6 (start),
March 31: extensions / open topics, TP (continuation),
**April 7: Oral presentations (50% of the grade)**.

TP sessions in Python, individually or in pairs, on

linear PDEs,
non-linear PDEs.

# Introduction

linear second-order parameter dependent problem

$$\begin{cases} -\nabla \cdot \big(a(\mu)\nabla u\big) = f & \text{dans } \Omega, \\ u = 0 & \text{sur } \partial\Omega. \end{cases}$$

$u(\mathbf{x};\mu) \in V$ : Unknowns,

$\mu \in \mathcal{G}$: Variable parameter,

$f \in L^2(\Omega)$,

$a : \mathcal{G} \times \Omega \to \mathcal{M}_d(\mathbb{R})$ is measurable, bounded, uniformly elliptic.

# Introduction

◇ **Main goal**

The objective of RBM is to find **very quickly accurate approximations** of parameter-dependent functions of the generic form

$$u : \Omega \times \mathcal{G} \to \mathbb{R},$$

- $\Omega \in \mathbb{R}^d$ : the spatial domain,
- $\mathcal{G} \subset \mathbb{R}^{N_p}$: the parameter domain, with $N_p$ the number of parameters.
  $\mu = (\mu_1, \ldots, \mu_{N_p}) \in \mathcal{G}$ : the varying parameter.

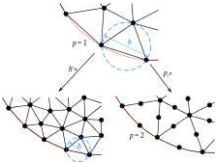
$$\mathcal{L}(\mu)(u(\mu)) = F(\mu), \text{ in } \Omega,$$
$$+ \text{ boundary conditions on } \partial\Omega.$$

# Introduction

Usually, classical methods such as

- ◇ Finite Difference Method (FDM)
- ◇ Finite Volume schemes (FV)
- ◇ Finite Element Method (FEM)

are used to provide an accurate approximation.
This consists in solving the problem in a subspace $V_h \subset V$, where $h$ is the mesh size.
Discrete approximations: $u_h \in V_h$.



# Introduction

## Finite Difference Methods

Mostly based on Taylor expansions of (smooth) solutions.
Cartesian geometry only (at least without any additional tools).
"Replace" derivatives by differential quotients:

$$\frac{\partial u}{\partial x} \rightsquigarrow \frac{u_{i+1} - u_i}{\Delta x}, \qquad \frac{\partial^2 u}{\partial x^2} \rightsquigarrow \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2}.$$

## Finite Volume Methods

Based on the conservation form of the PDE
Integrate the balance equation on each cell $\kappa$ and apply Stokes' formula:

$$\int_\kappa \text{source} = \sum_{\text{edges of } \kappa} \text{outward flux.}$$

Approximate each flux and write the discrete balance equation obtained.

## Finite Element Methods

Based on a variational formulation of the PDE.
Solve the formulation on a suitable finite-dimensional subspace of the energy space.
Piecewise polynomials: Finite Elements / Fourier-like basis: Spectral Methods

# Introduction

Let's get back to our sheep

linear second-order parameter dependent problem

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$u(\mathbf{x}; \boldsymbol{\mu}) \in V$ : Unknowns,

$\boldsymbol{\mu} \in \mathcal{G}$: Variable parameter,

# Introduction

We know how to solve this PDE with arbitrary accuracy, for instance using **Galerkin finite elements**.

Let $\{w_i\}_{1 \leq i \leq \mathcal{N}}$ be a finite element approximation basis (Lagrange $P^1$ hat functions). Then

$$u_h = \sum_{i=1}^{\mathcal{N}} u^i w_i$$

(where the $u^i$ are $\mathcal{N}$ real coefficients) defines a finite element approximation.

## Finite element linear system (Galerkin approximation)

The vector

$$\mathbf{u} = (u^1, \ldots, u^{\mathcal{N}})^T \in \mathbb{R}^{\mathcal{N}}$$

is the solution of a linear system

$$\mathbf{A}(\boldsymbol{\mu})\ \mathbf{u} = \mathbf{f}(\boldsymbol{\mu}),$$

where $\mathbf{A}(\boldsymbol{\mu}) \in \mathbb{R}^{\mathcal{N} \times \mathcal{N}}$ and $\mathbf{f}(\boldsymbol{\mu}) \in \mathbb{R}^{\mathcal{N}}$.

# Introduction

**In practice:** the finite element approximation basis $\{w_i\}_{1 \leq i \leq \mathcal{N}}$ relies on

- a mesh (partition of $\Omega$ into tetrahedra),
- for Lagrange $P^1$ finite elements, the number of degrees of freedom $\mathcal{N}$ equals the number of vertices.

## Convergence of the finite element method

Provided that *"sufficiently uniform"* meshes are used,

$$\|u_h - u\|_{H^1} = \mathcal{O}(h^s),$$

where $h$ denotes the mesh size and $s$ the order of convergence.

Arbitrary accuracy can be achieved. We speak of **High Fidelity (HF)** when the prescribed accuracy is considered satisfactory. In general, $\mathcal{N}$ is large.

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Cea' lemma

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V, \qquad \|u - I_h u\|_{H^1} \leq C h^{min(k,r)} |u|_{H^{r+1}}$$

where $h$ denotes the mesh size and $s$ the order of convergence.

Arbitrary accuracy can be achieved. We speak of **High Fidelity (HF)** when the prescribed accuracy is considered satisfactory. In general, $\mathcal{N}$ is large.

# Introduction

**Specific contexts:** requiring the evaluation of the PDE solution for a *very large number of parameter values*

- parametric studies,
- sensitivity analysis,
- uncertainty quantification,
- real-time simulations,
- etc.

Despite the use of HPC, associated computational costs might be prohibitive, especially for large-scale problems.

# Reduced basis methods

**How to reduce the computational costs of parameter-dependent problems?**

1 Introduction

2 Reduced Basis Methods

3 Galerkin projection

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# Reduced Basis Methods (RBM)

$PDE : \boldsymbol{\mu} \to u(\boldsymbol{\mu})$

$\boldsymbol{\mu} \in \mathcal{G}$ : Parameter

$u(\boldsymbol{\mu})$ : Solution

$$\begin{cases} -\nabla \cdot \big(a(\boldsymbol{\mu})\nabla u\big) = f(\boldsymbol{\mu}) & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

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$$u_h(\boldsymbol{\mu};\ \mathbf{x}) = \sum_{i=1}^{\mathcal{N}} u^i(\boldsymbol{\mu}) w_i(\mathbf{x}),$$

$$\begin{cases} \text{where } \mathbf{u}(\boldsymbol{\mu}) = (u^1(\boldsymbol{\mu}), \ldots, u^{\mathcal{N}}(\boldsymbol{\mu}))^T \in \mathbb{R}^{\mathcal{N}} \\ \text{ is the solution of a linear system} \\ \mathbf{A}(\boldsymbol{\mu})\mathbf{u}(\boldsymbol{\mu}) = \mathbf{f}(\boldsymbol{\mu}). \end{cases}$$

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**Aim of the reduced basis methods (RBM)**

Solve the PDE as quickly as possible when it has to be evaluated for many parameter values

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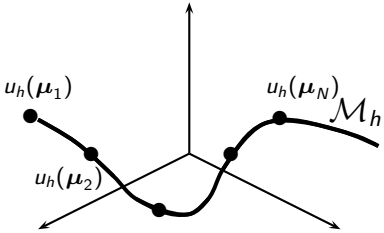
$u_h(\boldsymbol{\mu};\ \mathbf{x}) = \sum_{i=1}^{\mathcal{N}} u^i(\boldsymbol{\mu}) w_i(\mathbf{x})$,

**Solution manifold**

Solution manifold: $\mathcal{M} = \{u(\boldsymbol{\mu}) |\ \boldsymbol{\mu} \in \mathcal{G}\}$

HF solution manifold: $\mathcal{M}_h = \{u_h(\boldsymbol{\mu}) |\ \boldsymbol{\mu} \in\ \mathcal{G}\}$




Solution manifold

**Reduced space**

$V^N$ Reduced space:

$u_h(\boldsymbol{\mu}) \approx \sum_{i=1}^{N} \alpha_i(\boldsymbol{\mu}) u_h(\boldsymbol{\mu}_i)$

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**Does the manifolds have a favorable mathematical property that we could exploit?**

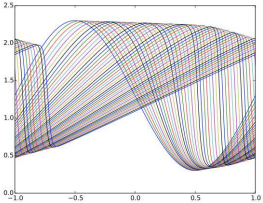


Fig. 1: Snapshots of the solution to the unsteady viscous Burger equation v
$\lambda = 1.3$, $\nu = 4$, $\varepsilon = 0.04$

**Reduced space**

$V^N$ Reduced space:

$u_h(\boldsymbol{\mu}) \simeq \sum_{i=1}^{N} \alpha_i(\boldsymbol{\mu}) u_h(\boldsymbol{\mu}_i)$

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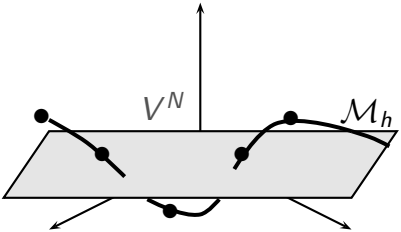


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$u_h(\boldsymbol{\mu}) \simeq \sum_{i=1}^{N} \alpha_i(\boldsymbol{\mu}) u_h(\boldsymbol{\mu}_i)$

$$V_N = \mathrm{Span}\{u_h(\boldsymbol{\mu}_1), u_h(\boldsymbol{\mu}_2), \ldots u_h(\boldsymbol{\mu}_N)\}$$

where $u_h(\boldsymbol{\mu}_1), \ldots, u_h(\boldsymbol{\mu}_N)$ = snapshots

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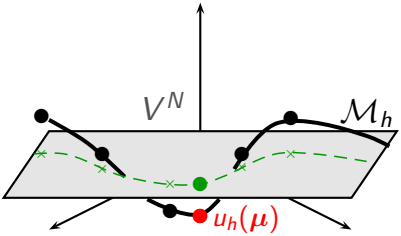


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# Examples

## Example 1

$PDE : \mu \to u(\mu)$

$\mu \in \mathcal{G} \subset \mathbb{R}$ : Parameter

$u(\mu)$ : Solution

$$\begin{cases} -\nabla \cdot \big(\mu \nabla u\big) = f(\mu) & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

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with $f(x) = \sin(2\pi x), \quad u(x;\ \mu) = \dfrac{1}{4\pi^2 \mu} \sin(2\pi x)$

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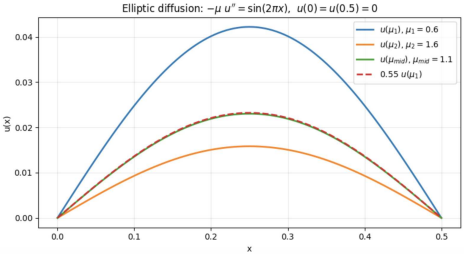
## Observation

With the solution for one parameter value $\mu_1$, we can approximate the solution of any other parameter value $\mu \in \mathcal{G}$



# Examples

$PDE : \boldsymbol{\mu} \to u(\boldsymbol{\mu})$

$\boldsymbol{\mu} \in \mathcal{G}$ : Parameter

$u(\boldsymbol{\mu})$ : Solution

**Question**

How many parameter values does one need to efficiently approximate the solution of any other parameter value?

$$\boxed{u(\boldsymbol{\mu}) \simeq \sum_{i=1}^{N} \alpha_i u(\boldsymbol{\mu}_i)}$$

# Examples

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# Examples

$PDE : \boldsymbol{\mu} \to u(\boldsymbol{\mu})$

$\boldsymbol{\mu} \in \mathcal{G}$ : Parameter

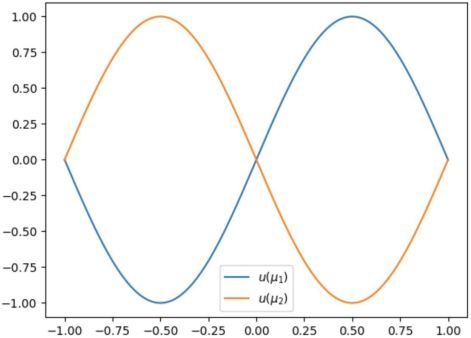
$u(\boldsymbol{\mu})$ : Solution

**Example 2**

$$\text{e.g. } u(x;\ \mu) = \sin(\pi(x - \mu))$$

$$u(x;\ \mu_1) = \sin(\pi x)$$

$$\begin{aligned} u(x;\ \mu_2) &= \sin(\pi(x-1)) \\ &= -\sin(\pi x) \end{aligned}$$



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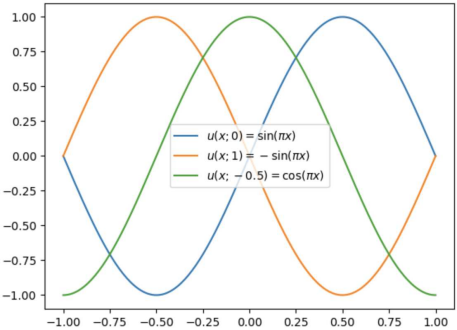

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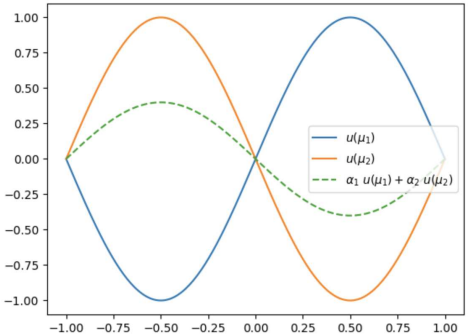

# Reduced Basis Methods (RBM)

- **Offline**
  $u(\mathbf{x}; \boldsymbol{\mu}_i)$
- **Online**
  $\alpha_i(\boldsymbol{\mu})$

$$u(\mathbf{x}; \boldsymbol{\mu}) \simeq \sum_{i=1}^{N} \alpha_i(\boldsymbol{\mu}) u(\mathbf{x}; \boldsymbol{\mu}_i)$$

# Reduced Basis Methods (RBM)

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  $u(\mathbf{x}; \boldsymbol{\mu}_i)$
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$$\cancel{u(\mathbf{x}; \boldsymbol{\mu}) \simeq \sum_{i=1}^{N} \alpha_i(\boldsymbol{\mu}) u(\mathbf{x}; \boldsymbol{\mu}_i)}$$

$$u(\mathbf{x}; \boldsymbol{\mu}) \simeq \sum_{i=1}^{N} \alpha_i(\boldsymbol{\mu}) \Phi_i(\mathbf{x})$$

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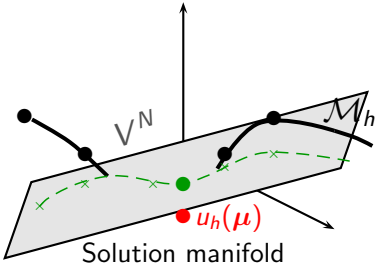
**Does the manifolds have a favorable mathematical property that we could exploit?**

**Reduced space**

$V^N$ Reduced space



# Reduced Basis Methods (RBM)

We look for the reduced **linear** space that best fit the solution manifold ...

# Reduced Basis Methods (RBM)

**Solution manifold**

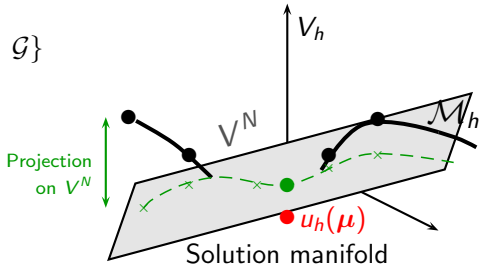
HF solution manifold: $\mathcal{M}_h = \{u_h(\boldsymbol{\mu}) |\ \mu \in \ \mathcal{G}\}$

**Reduced space**

$V^N$ Reduced space

Projection on $V^N$:

$$\boxed{\inf_{v_N \in V^N} \|u_h - v_N\|_{V_h}.}$$




We look for the reduced **linear** space that best fit the solution manifold:

$$\inf_{\substack{V^N \subset V_h \\ \dim(V^N)=N}} \sup_{u_h \in \mathcal{M}_h} \inf_{v_N \in V^N} \|u_h - v_N\|_{V_h}. \qquad \text{(M)}$$

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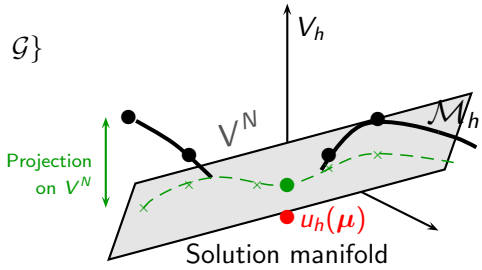


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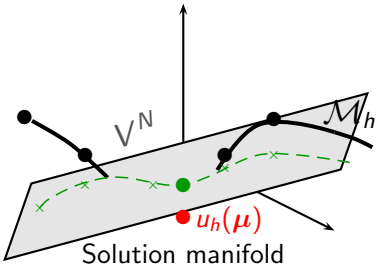
# Reduced Basis Methods (RBM)

Kolmogorov $N$-width:

$$\boxed{d_N(\mathcal{M}_h, V_h) = \inf_{\substack{V^N \subset V_h \\ \dim(V^N)=N}} \sup_{u_h \in \mathcal{M}_h} \inf_{v_N \in V^N} \|u_h - v_N\|_{V_h}.}$$




Favorable case:
Exponential decay

$$\exists, \tau, C > 0, \ \forall N > 1, \quad d_N(\mathcal{M}_h, V_h) \leq Ce^{-\tau N}.$$

**Few a priori results** (first result due to Maday, Patera, and Turinici (2002))
**Large number of a posteriori results**, covering a wide range of applications (elliptic, parabolic, and some—but not all—hyperbolic equations).

# Reduced Basis Methods

If we have $N$ small enough, such that $d_N(\mathcal{M}_h, V_h) = \varepsilon$, and if we find the minimizing space $V_N$, then

$\forall \boldsymbol{\mu} \in \mathcal{G}$, $\exists \boldsymbol{\alpha}(\boldsymbol{\mu}) = (\alpha_1(\boldsymbol{\mu}), \ldots, \alpha_N(\boldsymbol{\mu})) \in \mathbb{R}^N$ such that

$$\boxed{\|u_h(\boldsymbol{\mu}) - \sum_{n=1}^{N} \alpha_n(\boldsymbol{\mu})\Phi_n\|_V \leq \varepsilon,}$$

with $\{\Phi_n\}_{1 \leq n \leq N}$ a basis of $V^N$, called **the reduced basis**.

# Reduced Basis Methods

A reduced basis method is:

- **A reduced space spanned by a reduced basis:** how to build the infimum space $V^N$ (or to approach it)?
- **An approximation (in the reduced space):** how to find the $N$ coefficients $(\alpha_1(\boldsymbol{\mu}), \ldots, \alpha_N(\boldsymbol{\mu})) \in \mathbb{R}^N$ such that

$$u_N(\boldsymbol{\mu}) = \sum_{n=1}^{N} \alpha_n(\boldsymbol{\mu}) \Phi_n$$

  be the best approximation of $u(\boldsymbol{\mu})$ on $V^N$?

$u_N(\boldsymbol{\mu}) \in V^N$ is called **the reduced basis approximation.**

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Remarks: The best coefficients are the projection coefficients of $u_h(\boldsymbol{\mu})$:

$$\alpha_n(\boldsymbol{\mu}) = (u_h(\boldsymbol{\mu}), \Phi_n).$$

But of course, we don't want to compute $u_h(\boldsymbol{\mu})$ ...

# Reduced basis methods

- Galerkin-POD[a]
- Petrov-Galerkin RBM[b]
- EIM[c]
- POD-DL-ROM[d]
- 2-grid method[e]
- POD-I[f] (Reg, NN)
- PBDW[g]
- PGD[h]
- Surrogate problems[i]
- Operator inference[j]

...

[a] J.L. Lumley. *The structure of inhomogeneous turbulent flows.* 1967

[b] K. Carlberg, M. Barone, H. Antil. *Galerkin v. least-squares Petrov-Galerkin projection in nonlinear model reduction.* 2017

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# Reduced basis methods: https://reducedbasis.github.io

- Galerkin-POD[a]
- Petrov-Galerkin RBM[b]
- EIM[c]
- POD-DL-ROM[d]
- 2-grid method[e]
- POD-I[f] (Reg, NN)
- PBDW[g]
- PGD[h]
- Surrogate problems[i]
- Operator inference[j]

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# Reduced Basis Methods

- **Offline** Construction of a reduced space $V_N$ spanned by a reduced basis.
- **Online** Computation of the reduced coefficients $\boldsymbol{\alpha}$.

The optimal reduced space $V^N$ may not be found
Two main algorithms to find approximated reduced spaces: **the Proper Orthogonal Decomposition (POD)** or **greedy algorithms**.

In general, greedy algorithms are more efficient if they are combined with **aposteriori errors**.

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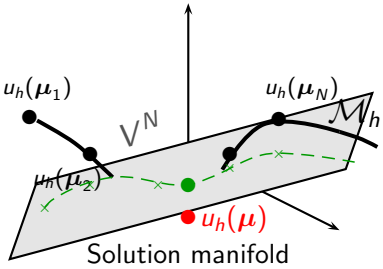
# Reduced Basis Methods

In both cases, the set of basis functions is derived from HF solutions for several well chosen parameter values, $\{u_h(\boldsymbol{\mu}_1), \ldots, u_h(\boldsymbol{\mu}_N)\}$, called **the snapshots**.



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- Classical discretization method (FEM/FV ...) with $\mathcal{N}$ huge!
  Question: How to decrease the High-Fidelity (HF) code execution runtimes when we need to solve the problem for **many different parameter values** while keeping the **same accuracy**?

  **ROM: Reduce the dimension of the algebraic system arizing from the discretization**
- RBM: The solution is obtained with a projection of the HF problem onto a reduced subspace.
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1 Introduction

2 Reduced Basis Methods

3 Galerkin projection

4 POD + TP VF

# Reduced basis Galerkin approximation

Suppose one have a reduced basis $(\Phi_i)_{i=1,\dots,N}$ of $V_N$
We seek $\alpha_i(\boldsymbol{\mu})$ to get the approximation

$$u_N(\mathbf{x};\boldsymbol{\mu}) = \sum_{i=1}^{N} \alpha_i(\boldsymbol{\mu})\Phi_i(\mathbf{x})$$

# Reduced basis Galerkin approximation

Let's get back to our sheep

linear second-order parameter dependent problem

$$\begin{cases} -\nabla \cdot \big(a(\boldsymbol{\mu})\nabla u\big) = f(\boldsymbol{\mu}) & \text{dans } \Omega, \\ u = 0 & \text{sur } \partial\Omega. \end{cases}$$

$u(\mathbf{x}; \boldsymbol{\mu}) \in V$ : Unknowns,

$\boldsymbol{\mu} \in \mathcal{G}$: Variable parameter,

## Exercise

1. Write the weak formulation of the problem.
2. Prove that the problem is well-posed under some conditions.

# Reduced basis Galerkin approximation

Context: $V_N \subset V$ is a subspace of dimension $N$.

**Definition**

The Galerkin reduced-basis approximation is the function $u_N(\boldsymbol{\mu}) \in V_N$ satisfying the weak form

$$\forall v_N \in V_N, \qquad a\left(u_N(\boldsymbol{\mu}), v_N; \boldsymbol{\mu}\right) = \ell\left(v_N; \boldsymbol{\mu}\right).$$

## Exercise

Show that if $a(\cdot, \cdot; \boldsymbol{\mu}) : V \times V \to \mathbb{R}$ is coercive, then the Galerkin reduced-basis approximation exists and is unique.

# Reduced basis Galerkin approximation

Assume the weak formulation of the HF problem yields the discretized system

$$\mathbf{A}(\boldsymbol{\mu})\mathbf{u}(\boldsymbol{\mu}) = \mathbf{l}(\boldsymbol{\mu})$$

then

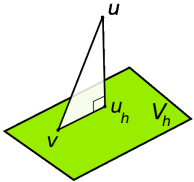
$$a\left(u_N(\boldsymbol{\mu}), v_N; \boldsymbol{\mu}\right) = \ell\left(v_N; \boldsymbol{\mu}\right)$$

gives a new system to solved:

$$\boxed{\mathbf{P}^T\mathbf{A}(\boldsymbol{\mu})\mathbf{P}\ \boldsymbol{\alpha}(\boldsymbol{\mu}) = \mathbf{P}^T\mathbf{l}(\boldsymbol{\mu}),} \qquad \text{(G-RB)}$$

where $\mathbf{P} \in \mathbb{R}^{\mathcal{N}\times N}$.

# Reduced basis Galerkin approximation




**G-RB proof**

## FEM recall: Standard FEM Galerkin condition

The error $e_h$ is orthogonal to the test space $V_h$ in the energy inner product induced by $a(\cdot,\cdot)$ :

$$\boxed{a(e_h, v_h) = a(u - u_h, v_h) = 0, \ \forall v_h \in V_h.}$$

Consider the weak residual $r(v_h) = \ell(v_h) - a(u_h, v_h)$. Then

$$r(v_h) = a(u, v_h) - a(u_h, v_h) = a(e_h, v_h) = 0, \ \forall v_h \in V_h.$$

In other words, the residual vanishes on the test space.

# Reduced basis Galerkin approximation

Let $\mathbf{P} = [\Phi_1, \ldots, \Phi_N] \in \mathbb{R}^{N \times \mathcal{N}}$, i.e. $\Phi_j = \sum_{i=1}^{\mathcal{N}} P_{ij} w_i$ and $P = (P_{ij}) \in \mathbb{R}^{N \times \mathcal{N}}$.

## RB Galerkin condition

1. $u_N(\mu) = \sum_{n=1}^{N} \alpha_n(\mu)\Phi_n = \mathbf{P}\alpha(\mu)$, $\alpha = \alpha(\mu) \in \mathbb{R}^N$, $\mathbf{r}(\alpha) = \mathbf{l}(\mu) - \mathbf{A}(\mu)(\mathbf{P}\alpha)$.

2. With HF basis $\{w_i\}_{i=1}^{\mathcal{N}}$, we know that:

$$\ell(v_N) = \mathbf{v_N}^T \mathbf{l}, \quad a(u_N, v_N) = \mathbf{v_N}^T \mathbf{A} \mathbf{u_N}, \ \forall v_N \in V_N \subset V_h,$$

with $\mathbf{A}_{ij} = a(w_j, w_i)$, $\mathbf{l}_i = \ell(w_i)$
So $r(v_N) = \mathbf{v_N}^T \mathbf{r}$.

3. The residual is orthogonal to the test space $V_N$ : $r(v_N) = 0$, $\forall v_N \in V_N$. Thus, $\forall v_N \in V_N$, $\mathbf{v_N}^T \mathbf{r}(\alpha) = 0$.

4. But $v_N \in V_N$, so $v_N = \mathbf{P}\beta$ with $\beta \in \mathbb{R}^N$, and $(\mathbf{P}\beta)^T(\mathbf{l}(\mu) - \mathbf{A}(\mu)(\mathbf{P}\alpha(\mu))) = 0$.
True for all $\beta \in \mathbb{R}^N$: $\mathbf{P}^T(\mathbf{l}(\mu) - \mathbf{A}(\mu)\mathbf{P}\alpha(\mu)) = 0$.

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Interpretation: With the standard RB Galerkin projection, we look for $\boldsymbol{\alpha}$ that minimizes the error in the energy norm.

$$u_N = \arg \min_{v_N \in V_N} \|u - v_N\|_a$$

Now, we get a system where the inversion cost is in $\mathcal{O}(N^3)$ since dimensions : $\mathbf{P}^T\mathbf{A}(\boldsymbol{\mu})\mathbf{P} \in \mathbb{R}^{N\times N}$ and $\mathbf{P}^T b \in \mathbb{R}^N$!

*Remarks*

- Other projections can be considered (e.g. Petrov-Galerkin projection $\mathbf{W}^T\mathbf{A}\mathbf{P}\boldsymbol{\alpha} = \mathbf{W}^T\mathbf{I}$ with a test basis $\mathbf{W} \neq \mathbf{P}$).
- In (G-RB), we first need to assemble $\mathbf{A}(\boldsymbol{\mu}) \in \mathbb{R}^{\mathcal{N}\times\mathcal{N}}$.

## Exercise

Consider the PDE model and derive an offline/online strategy to overcome the complexity of the assembly process.

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# Reduced basis Galerkin approximation

**Affine operator**
**definition**

the bilinear for $a(\cdot,\cdot;\ \boldsymbol{\mu}) : V \times V \to \mathbb{R}$ is affine (with respect to the parameter) if there exist

- ◇ $Q^a$ functions $\theta_q^a : \mathcal{G} \to \mathbb{R},\ 1 \leq q \leq Q^a$ bounded (i.e. $\theta_q^a(\mathcal{G})$),
- ◇ $Q^a$ functions $a_q : V \times V \to \mathbb{R},\ 1 \leq q \leq Q^a$

such that

$$\forall \boldsymbol{\mu} \in \mathcal{G},\ \forall (u,v) \in V \times V, a(u,v;\ \boldsymbol{\mu}) = \sum_{q=1}^{Q^a} \theta_q^a(\boldsymbol{\mu})\ a_q(u,v).$$

$$\boxed{\mathbf{A}(\boldsymbol{\mu}) = \sum_{q=1}^{Q^a} \theta_q^a(\boldsymbol{\mu})\ \mathbf{A}_q}$$

# Reduced basis Galerkin approximation

**And surprisingly for l ...**
**definition**
the linear for $l(\cdot;\ \boldsymbol{\mu}) : V'$ is affine if there exist

- ◇ $Q^l$ functions $\theta_l^a : \mathcal{G} \to \mathbb{R},\ 1 \leq q \leq Q^l$ bounded (i.e. $\theta_q^l(\mathcal{G})$),
- ◇ $Q^l$ functions $l_q : V',\ 1 \leq q \leq Q^l$

such that

$$\forall \boldsymbol{\mu} \in \mathcal{G},\ \forall v \in V, a(v;\ \boldsymbol{\mu}) = \sum_{q=1}^{Q^l} \theta_q^l(\boldsymbol{\mu}) l_q(v).$$

$$\boxed{\mathbf{l}(\boldsymbol{\mu}) = \sum_{q=1}^{Q^l} \theta_q^l(\boldsymbol{\mu})\ \mathbf{l}_q}$$

# Reduced basis Galerkin approximation

Assembling cost with the affine operators:
$\mathcal{O}(N^2 Q^a + NQ^l)$ with

$$\mathbf{P}^T\mathbf{A}(\boldsymbol{\mu})\mathbf{P} = \sum_{q=1}^{Q^a} \theta_q^a(\boldsymbol{\mu}) \underbrace{\mathbf{P}^T\mathbf{A}_q\mathbf{P}}_{\text{precomputed offline}}, \quad \mathbf{P}^T\mathbf{l}(\boldsymbol{\mu}) = \sum_{q=1}^{Q^l} \theta_q^l(\boldsymbol{\mu}) \underbrace{\mathbf{P}^T\mathbf{l}_q}_{\text{precomputed offline}}.$$

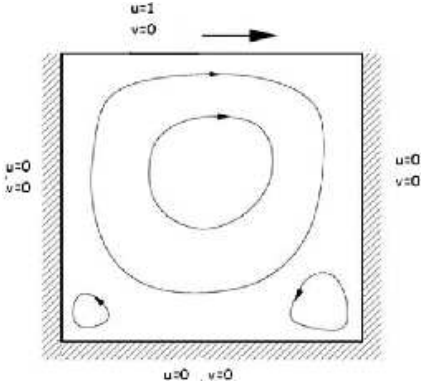
# Reduced basis Galerkin approximation

Visit `https://reducedbasis.github.io/docs/pod/` for the reduced basis Galerkin approximation of the Navier-Stokes problem.

# Navier-Stokes

## 2D lid driven cavity problem

Stationary Navier-Stokes equation in the unit square (+ Dirichlet boundary cdts)




2D steady Navier-stokes equation:

$$
\begin{aligned}
(\mathbf{u}\cdot\nabla)\mathbf{u} - \mu\Delta\mathbf{u} + \nabla p &= 0, \text{ on } \Omega,\\
\nabla\cdot\mathbf{u} &= 0, \text{ on } \Omega,\\
\mathbf{u}_{|\Gamma_{up}} &= (1,0),\\
\mathbf{u}_{|\partial\Omega\setminus\Gamma_{up}} &= (0,0),
\end{aligned}
$$

where $\mathbf{u} = (u, v)$ is the velocity,
$p$ is the pressure,
$\Gamma_{up}$ is the upper border,
$\mu = 1/Re$, and $Re$ is the Reynolds number.

# POD Galerkin

Instead of using $\mathbf{u}$ as before, we replace it by $\overline{\mathbf{u}} + \sum_{j=1}^{N} \alpha_j^k \Phi_j$
On the **2D lid driven cavity problem**, we obtain:

$$\mathcal{A}_i + \sum_{j=1}^{N} \mathcal{B}_{ij}\alpha_j^k + \sum_{j=1}^{N}\sum_{r=1}^{N} \mathcal{C}_{ijr}\alpha_j^k \alpha_r^{k-1} = 0,$$

where $\mathcal{A} \in \mathbb{R}^N, \mathcal{B} \in \mathbb{R}^{N\times N}$ and $\mathcal{C}_{ijr} = \mathbb{R}^{N\times N^2}$. e.g. $\mathcal{A}_i = \underbrace{\mu(\nabla\overline{\mathbf{u}}, \nabla\Phi_i)}_{A1} + \underbrace{(\overline{\mathbf{u}}\cdot\nabla\overline{\mathbf{u}}, \Phi_i)}_{A2}$

## Online stage

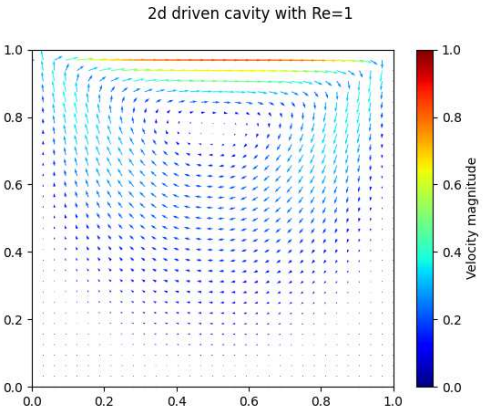
Input: New parameter of interest $\mu$

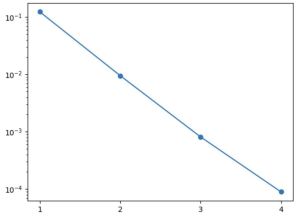
$$M_{ij} = \mathcal{B}_{ij} + \sum_{k=1}^{N} \mathcal{C}_{ijl}\alpha_l^{k-1}$$

Output:

$$\mathbf{M}\alpha^k(\mu) = \boldsymbol{\mathcal{A}}, \qquad \mathbf{u}^N(x;\mu) = \overline{\mathbf{u}} + \sum_{j=1}^{N} \alpha_j(\mu)\Phi_h^j(x).$$

# POD Galerkin




$1-$RIC (logscale)

Eigenvalues: [2.2E-02, 2.9E-03, 2.2E-04, 1.9E-05,...]

# Sum up

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- Classical discretization method (FEM/FV ...) with $\mathcal{N}$ huge!
  Question: How to decrease the High-Fidelity (HF) code execution runtimes when we need to solve the problem for **many different parameter values** while keeping the **same accuracy**?

  **ROM: Reduce the dimension of the algebraic system arizing from the discretization**

- RBM: The solution is obtained with a projection of the HF problem onto a reduced subspace.
- How to obtain the reduced basis?
  POD (TP 1-2-3) /Greedy algorithms (TP4)
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# Reduced bases on reduced basis methods

Purpose:
Find **very quickly** for a new parameter $\boldsymbol{\mu} \in \mathcal{G}$ an approximation on $V^N$!

$$\boxed{\text{costly FEM/FV}} \Rightarrow \underbrace{\boxed{\begin{array}{c}\text{OFFLINE:}\\ \text{Construction of } V^N \\ \text{from snapshots}\end{array}}}_{\text{POD/Greedy}} \Rightarrow \underbrace{\boxed{\begin{array}{c}\text{ONLINE:}\\ \text{Computation of}\\ \text{the coefficients}\\ (\alpha_i)_{i=1,\dots,N}\end{array}}}_{\text{Galerkin projections}}$$

1 Introduction

2 Reduced Basis Methods

3 Galerkin projection

4 POD + TP VF

# POD: Continuous version

We want to approximate $u(\mathbf{x}, \boldsymbol{\mu})$ by $\sum_{k=1}^{N} a_k(\boldsymbol{\mu})\Phi_k(\mathbf{x})$, with orthonormalized $\Phi_k$, where $a_k(\boldsymbol{\mu}) = (u_h(\boldsymbol{\mu}), \Phi_k)$.

# POD: Continuous version

$u - P_{V^N} u \perp V^N$

$(u(\boldsymbol{\mu}) - \sum_{i=1}^{N} \alpha_i(\boldsymbol{\mu})\Phi_i, \Phi_j) = 0, \; j = 1, \dots N$

$\sum_{i=1}^{N} (\Phi_i, \Phi_j)\alpha_i(\boldsymbol{\mu}) = (u(\boldsymbol{\mu}), \Phi_i)$

$\Leftrightarrow G\boldsymbol{\alpha}(\boldsymbol{\mu}) = b$

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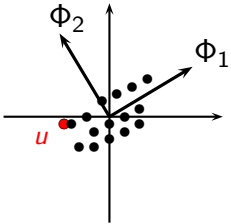
# POD: Continuous version

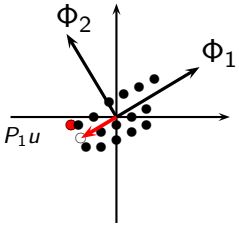
We want to approximate $u(\mathbf{x}, \boldsymbol{\mu})$ by $\sum_{k=1}^{N} a_k(\boldsymbol{\mu})\Phi_k(\mathbf{x})$.

Let us consider $\boldsymbol{\mu}$ a random variable and $u$ centered ($\mathbb{E}_{\boldsymbol{\mu}}[u] = 0$).
POD = PCA: We want to find the axes that best represent the data!




$$\boxed{\min_{\|\Phi_i\|=1} \mathbb{E}[\|u - \sum_{k=1}^{N} a_k(\boldsymbol{\mu})\ \Phi_k\|^2].}$$

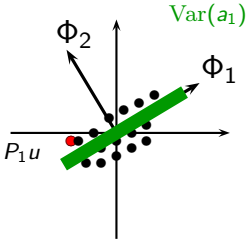
# POD: Continuous version




Snapshot projection over $\Phi_1$: $(u, \Phi_1)\Phi_1$

# POD: Continuous version

**How do we find the reduced basis?**




$$Var(a_1) = \mathbb{E}[a_1^2] - (\mathbb{E}[a_1])^2 = \mathbb{E}[a_1^2]$$

$$a_1 = (u, \Phi_1), \ \|\Phi_1\| = 1.$$

$$\min_{\|\Phi_1\|=1} \mathbb{E}[\|u - (u, \Phi_1)\Phi_1\|^2] \text{ or } \max_{\|\Phi_1\|=1} \mathbb{E}[|(u, \Phi_1)|^2].$$

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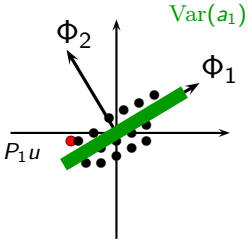


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# POD: Continuous version

We want to approximate $u(\mathbf{x}, \boldsymbol{\mu})$ by $\sum_{k=1}^{N} a_k(\boldsymbol{\mu})\Phi_k(\mathbf{x})$,
with $\Phi_k \in L^2(\Omega)$ + orthonormality $\int_\Omega \Phi_i(\mathbf{x})\Phi_j(\mathbf{x})\, d\mathbf{x} = \delta_{ij}$.

We look for $\Phi_i,\ i = 1, \ldots, N$ that maximize the average projection:

$$\max_{\|\Phi_i\|=1} \mathbb{E}[|(u, \Phi_i)|^2].$$

With $\Phi_1 \in V$ where $V$ is a real Hilbert space, one can show that this maximization problem is equivalent to a spectral problem

$$\boxed{C\Phi_1 = \lambda_1 \Phi_1}$$

(assuming that the dominated convergence theorem can be applied).

# POD: Continuous version

**Proof**

Lagrangian: $J[\Phi_1] = \mathbb{E}[|(u, \Phi_1)|^2] - \lambda(\|\Phi_1\|^2 - 1)$.

Directional derivative (with a small variation $\Psi$):

$J[\Phi + \delta\Psi] = \mathbb{E}[|(u, \Phi + \delta\Psi)|^2] - \lambda(\|\Phi + \delta\Psi\|^2 - 1)$.

$\frac{d}{d\delta} J[\Phi + \delta\Psi]_{|\delta=0} = 0$.

- $(u, \Phi + \delta\Psi) = (u, \Phi) + \delta(u, \Psi)$,
  so $(u, \Phi + \delta\Psi)^2 = (u, \Phi)^2 + 2\delta(u, \Phi)(u, \Psi) + O(\delta^2)$
  and with average:
  $\mathbb{E}[(u, \Phi + \delta\Psi)^2] = \mathbb{E}[(u, \Phi)^2] + 2\delta\mathbb{E}[(u, \Phi)(u, \Psi)] + O(\delta^2)$
  So $\boxed{\frac{d}{d\delta}\mathbb{E}[(u, \Phi + \delta\Psi)^2]_{|\delta=0} = 2\mathbb{E}[(u, \Phi)(u, \Psi)]}$
- $(\Phi + \delta\Psi, \Phi + \delta\Psi) = (\Phi, \Phi) + 2\delta(\Phi, \Psi) + O(\delta^2)$, so
  $\boxed{\frac{d}{d\delta}(\Phi + \delta\Psi, \Phi + \delta\Psi)_{|\delta=0} = 2(\Phi, \Psi).}$

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# POD: Continuous version

**Proof**

Altogether, we obtain:

$$\boxed{\mathbb{E}[(u,\Phi)(u,\Psi)] - \lambda(\Phi,\Psi) = 0}$$

Let $C$ be such that $(C\Phi,\Psi) = \mathbb{E}[(u,\Phi)(u,\Psi)]$. Then

$$(C\Phi,\Psi) = \lambda(\Phi,\Psi), \forall \Psi.$$

Since true for all $\Psi$, we obtain:

$$\boxed{\mathbb{E}[(u,\Phi)u] = \lambda\Phi, \text{ i. e. } C\Phi = \lambda\Phi}.$$

We can prove that $C$ is a positive linear compact self-adjoint operator: the problem is well-posed.

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# Snapshots POD

**Spectral theorem (Compact Self-Adjoint Operator)**
Let $V$ be a separable Hilbert space and let $C : V \to V$ be a compact, positive, self-adjoint operator. Then there exists an orthonormal basis $\{\Phi_n\}_{n\in\mathbb{N}}$ of $V$ and a sequence of real numbers $(\lambda_n)_{n\in\mathbb{N}} \geq 0$ such that, for all $n \in \mathbb{N}$,

$$C\Phi_n = \lambda_n \Phi_n.$$

Moreover, $\lim_{n\to\infty} \lambda_n = 0$.

**Proof**

- Assume that there exists $\lambda_n$ such that $\lim_{n\to\infty} \lambda_n \neq 0$.
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  Then there exists $\varepsilon > 0$ such that $|\lambda_{n_k}| \geq \varepsilon$.
- Let $\Phi_{n_k}$ be the corresponding orthonormal eigenvectors: $C\Phi_{n_k} = \lambda_{n_k}\Phi_{n_k}$.
  $\Phi_{n_k}$ bounded: $\|\Phi_{n_k}\| = 1$
  $C$ compact $\Rightarrow$ $(C\Phi_{n_k})$ admit a convergent subsequence

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  $\Phi_{n_k}$ bounded: $\|\Phi_{n_k}\| = 1$
  $C$ compact $\Rightarrow (C\Phi_{n_k})$ admit a convergent subsequence
- $(\Phi_n)$ orthonormal $\Rightarrow \|\Phi_n - \Phi_m\| = \sqrt{2} \quad$ for $n \neq m$.
  For the subsequence: $\|C\Phi_{n_{k'_k}} - C\Phi_{m_{k'_k}}\| = \|\lambda_{n_{k'_k}}\Phi_{n_{k'_k}} - \lambda_{m_{k'_k}}\Phi_{m_{k'_k}}\| \nrightarrow 0 \quad$ if $|\lambda_{n_k}| \geq \varepsilon$.

This contradicts the compactness of $C$.

# Snapshot POD

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Let $V$ be a separable Hilbert space and let $C : V \to V$ be a compact, positive, self-adjoint operator. Then there exists an orthonormal basis $\{\Phi_n\}_{n\in\mathbb{N}}$ of $V$ and a sequence of real positive numbers $(\lambda_n)_{n\in\mathbb{N}}$ such that, for all $n \in \mathbb{N}$,

$$C\Phi_n = \lambda_n \Phi_n.$$

Moreover, $\lim_{n\to\infty} \lambda_n = 0$.

$$\boxed{\mathbb{E}[(u,\Phi)u] = \lambda\Phi, \text{ i. e. } C\Phi = \lambda\Phi}.$$

One can prove that $C$ is a positive linear compact self-adjoint operator:
one unique solution equal to the largest eigenvalue of the problem!
$(C\Phi,\Phi) = \mathbb{E}[|(u,\Phi)|^2] = \lambda$
$\max\limits_{\|\Phi_1\|=1} \mathbb{E}[|(u,\Phi_1)|^2] = \lambda_1$

# POD energy error

$\lambda_1 >= \lambda_2 >= \ldots >= 0$

$$\boxed{\mathbb{E}[\|u - P_N u\|^2] = \sum_{k>N} \mathbb{E}[a_k^2] = \sum_{k>N} \lambda_k}$$

**Proof:**
Since $\{\Phi_k\}$ is an orthonormal basis of $V$, $u(\boldsymbol{\mu}) = \sum_{k=1}^{\infty} a_k(\boldsymbol{\mu})\Phi_k$.

$$\mathbb{E}[\|u - P_N u\|^2] = \mathbb{E}[\|\sum_{k=N}^{\infty} a_k \Phi_k\|^2] = \sum_{k>N} \mathbb{E}[a_k^2] = \sum_{k>N} \lambda_k$$

In fact, one can show that the more regularizing the operator $C$ is, the faster its eigenvalues decay!

# Link between regularity and eigenvalues

One can show that the more regularizing the operator $C$ is, the faster its eigenvalues decay!

## Exercise

Let $A : L^2(0,1) \to L^2(0,1)$ be defined by $(Ax)(t) = \int_0^t x(s)\,ds$.

**1.** Determine the adjoint operator $A^*$.

**2.** Define $C = A^*A$. Show that $C$ is a compact, positive, and self-adjoint operator on $L^2(0,1)$.

**3.** Let $\sigma$ denote a singular value of $A$, and let $x \neq 0$ satisfy $Cx = A^*Ax = \sigma^2 x$.

(a) Show that $A^*Ax$ is twice differentiable and compute $(A^*Ax)''(t)$.

(b) Deduce that $x$ satisfies the differential equation $x(t) + \sigma^2 x''(t) = 0$.

**4.** Using the appropriate boundary conditions, determine the general form of the eigenfunctions $x(t)$.

**5.** Deduce the explicit expression of the singular values $\sigma_n$ of $A$ and determine their asymptotic behaviour as $n \to \infty$.

# Link between regularity and eigenvalues

## Solution

Let $A : L^2(0,1) \to L^2(0,1)$ be defined by $(Ax)(t) = \int_0^t x(s)\,ds$.

**1. Adjoint $A^*$.** For $x, y \in L^2(0,1)$,
$(Ax, y) = \int_0^1 \left(\int_0^t x(s)\,ds\right) y(t)\,dt = \int_0^1 x(s) \left(\int_s^1 y(t)\,dt\right)\,ds = (x, A^*y)$,

**2.**

*Self-adjointness:* $C = A^*A$ is self-adjoint since $(A^*A)^* = A^*(A^*)^* = A^*A$.

*Positivity:* $(Cx, x) = (A^*Ax, x) = (Ax, Ax) = \|Ax\|_{L^2}^2 \geq 0$.

*Compactness:* $A$ maps $L^2(0,1)$ continuously into $H^1(0,1)$, since $Ax \in H^1(0,1)$ and

$$(Ax)' = x \quad \text{in } L^2(0,1), \qquad \|Ax\|_{H^1}^2 = \|Ax\|_{L^2}^2 + \|x\|_{L^2}^2 \leq C\|x\|_{L^2}^2.$$

The embedding $H^1(0,1) \hookrightarrow L^2(0,1)$ is compact (Rellich), hence $A$ is compact and consequently $C = A^*A$ is compact.

# Link between regularity and eigenvalues

## Solution

**3. Spectral problem.** Let $\sigma$ be a singular value of $A$ and let $x \neq 0$ satisfy $Cx = A^*Ax = \sigma^2 x$.
Set $y = A^*Ax$. Since $(A^*y)(t) = \int_t^1 y(s)\,ds$, $(A^*y)'(t) = -y(t)$ (p.p).
Hence $(A^*Ax)'(t) = -(Ax)(t)$ (p.p), $(A^*Ax)''(t) = -x(t)$
On the other hand, $A^*Ax = \sigma^2 x$ implies $(A^*Ax)'' = \sigma^2 x''$, thus
$\sigma^2 x''(t) = -x(t) \iff x(t) + \sigma^2 x''(t) = 0.$

**Boundary conditions.** From $A^*Ax = \sigma^2 x$:
at $t = 1$, $(A^*Ax)(1) = \int_1^1 (Ax)(s)\,ds = 0 = \sigma^2 x(1) \implies x(1) = 0$. Also,
$(A^*Ax)'(0) = -(Ax)(0) = 0 = \sigma^2 x'(0) \implies x'(0) = 0.$

**4. Eigenfunctions.** Let $\omega = \sigma^{-1}$. The ODE becomes $x'' + \omega^2 x = 0$, so
$x(t) = A\cos(\omega t) + B\sin(\omega t)$. The condition $x'(0) = 0$ gives $B = 0$. Then $x(1) = 0$ yields
$A\cos(\omega) = 0 \iff \omega = \frac{\pi}{2} + n\pi, \qquad n \in \mathbb{N}$. Thus the (normalized) eigenfunctions are

$$x_n(t) = A\cos\left(\left(n + \tfrac{1}{2}\right)\pi\, t\right).$$

# Link between regularity and eigenvalues

## Solution

**5. Singular values and asymptotics.** Since $\omega_n = \sigma_n^{-1} = \left(n + \frac{1}{2}\right)\pi$, we obtain $\sigma_n = \frac{1}{\left(n+\frac{1}{2}\right)\pi} = \frac{2}{(2n+1)\pi}$, $\quad n \in \mathbb{N}$. Therefore, $\sigma_n \sim \frac{1}{\pi n}$ as $n \to \infty$. The eigenvalues of $C = A^* A$ are $\mu_n = \sigma_n^2$, hence $\mu_n = \frac{1}{\left(\left(n+\frac{1}{2}\right)\pi\right)^2} = \frac{4}{(2n+1)^2\pi^2} \sim \frac{1}{\pi^2 n^2}$.

# Link between regularity and eigenvalues

One can show that the more regularizing the operator $C$ is, the faster its eigenvalues decay!

## Exercise: Transport equation

Let $T_h : L^2(0,1) \to L^2(0,1)$ be the transport operator defined by $(T_h x)(t) = x(t-h)$, with periodic boundary conditions on $(0,1)$ and a fixed shift $h \in (0,1)$.

1. Show that $T_h$ is a bounded linear operator on $L^2(0,1)$ and that $\|T_h x\|_{L^2} = \|x\|_{L^2}$.
2. Compute the adjoint $T_h$ and show that $T_h^* = T_{-h}$.
3. Show that $T_h^* T_h = I$, where $I$ is the identity operator.
4. Deduce that the singular values of $T_h$ satisfy $\sigma_n = 1 \quad$ for all $n$.
5. Conclude that $T_h$ is not compact and does not regularize.

# Link between regularity and eigenvalues

## Solution

**1.** Linearity is immediate. Moreover, $\|T_h x\|^2_{L^2(0,1)} = \int_0^1 |x(t-h)|^2\,dt = \int_0^1 |x(t)|^2 dt$, due to periodicity. Hence $\|T_h x\|_2 = \|x\|_2$ for all $x$, so $T_h$ is bounded and $\|T_h\| = 1$.

**2. Adjoint.** For $x, y \in L^2(0,1)$, $(T_h x, y) = \int_0^1 x(t-h)\,\overline{y(t)}\,dt$.
Let $u = t - h$. Using periodicity, $\int_0^1 x(t-h)\,\overline{y(t)}\,dt = \int_0^1 x(u)\,\overline{y(u+h)}\,du = (x,\ T_{-h}y)$, since $(T_{-h}y)(u) = y(u+h)$. Therefore $T_h^* = T_{-h}$.

**3.** Using $T_h^* = T_{-h}$, for any $x$,
$(T_h^* T_h x)(t) = (T_{-h} T_h x)(t) = (T_h x)(t+h) = x((t+h)-h) = x(t)$. Hence $T_h^* T_h = I$.

**4. Singular values.** By definition, the singular values are the square roots of the eigenvalues of $T_h^* T_h$ (counted with multiplicity). Since $T_h^* T_h = I$, all eigenvalues equal 1, with infinite multiplicity. Thus $\sigma_n(T_h) = 1, \qquad \forall n \in \mathbb{N}$.

# Link between regularity and eigenvalues

## Solution

**5.**

- Let $e_k(t) = e^{2\pi ikt}$. Then $\{e_k\}$ orthonormal family
- $T_h\ e_k(t) = e_k(t-h) = e^{-2\pi ikh}\ e_k(t)$. So, $\{T_h e_k\}$ is also orthonormal: no converging subsequence.
- Moreover, $T_h$ is an isometry (unitary), so it does not attenuate high frequencies and therefore does not regularize.

# Link between regularity and eigenvalues

Thus $\sigma = 1$ No decrease! **No efficient basis functions for transport and advection-dominated regime**

- No regularity gain
- No compacity
- No decrease in POD eigenvalues
- No efficient basis functions

What can we do in that case?

- shifted POD
- transport maps
- nonlinear reduced bases
- POD + autoencoders

# Link between regularity and eigenvalues

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# Link between regularity and eigenvalues

Thus $\sigma = 1$ No decrease!

- No regularity gain
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**What can we do in that case?**

**Transport:**

$$\begin{cases} \text{shifted POD: } u(x,t) = \sum_k \sum_n a_{k,n}(t)\Phi_{k,n}(\mathbf{x} - \mathbf{s}_k(t)) \\ \text{transport maps: } u(x,t) = \sum_n a_n(t)\Phi_n(T_t(\mathbf{x})) \end{cases}$$

**Nonlinear dynamics:**

$$\begin{cases} \text{nonlinear reduced bases: } = \mathrm{F(u)} \leftrightarrow \dot{a} = Aa + H(a \otimes a) \\ \text{POD+autoencoders [1]} (\mathrm{t}, \mu) \rightarrow NN \rightarrow a(t,\mu) \rightarrow u \simeq Va \end{cases}$$

[1] POD_DL_ROM: `https://reducedbasis.github.io/docs/poddlrom/`

# Sheep model

Let's get back to our sheep

linear second-order parameter dependent problem

$$\begin{cases} -\nabla \cdot \big(a(\boldsymbol{\mu})\nabla u\big) = f(\boldsymbol{\mu}) & \text{dans } \Omega, \\ u = 0 & \text{sur } \partial\Omega. \end{cases}$$

$u(\mathbf{x}; \boldsymbol{\mu}) \in V$ : Unknowns,

$\boldsymbol{\mu} \in \mathcal{G}$: Variable parameter,

**How do we construct the reduced basis with the POD in practice?**

# Snapshot POD

$$\boxed{\mathbb{E}[(u,\Phi)u] = \lambda\Phi, \text{ i. e. } C\Phi = \lambda\Phi}.$$

$\Leftrightarrow \int_\Omega R(\mathbf{x},\mathbf{x}')\ \Phi(\mathbf{x}')d\mathbf{x}' = \lambda\Phi(\mathbf{x})$, with $R(\mathbf{x},\mathbf{x}') = \mathbb{E}[u(\mathbf{x})u(\mathbf{x}')]$, which is the spatial continuous covariance operator.

# Snapshot POD

The space of the modes is contained into the snapshots' span.

$$R(\mathbf{x},\mathbf{x}') = \mathbb{E}[u(\mathbf{x})u(\mathbf{x}')] \simeq \frac{1}{N_{train}} \sum_{k=1}^{N_{train}} u(\mathbf{x},\boldsymbol{\mu}_k)u(\mathbf{x}',\boldsymbol{\mu}_k).$$

$$\boxed{\mathbb{E}[\underbrace{(u,\Phi)}_{a}\, u] = \lambda\Phi, \text{ i. e. } C\Phi = \lambda\Phi}.$$

$$\text{so } \frac{1}{N_{train}} \sum_{k=1}^{N_{train}} u(\mathbf{x},\boldsymbol{\mu}_k) \underbrace{\int_\Omega u(\mathbf{x}',\boldsymbol{\mu}_k)\Phi(\mathbf{x}')d\mathbf{x}'}_{a_k=(u(\boldsymbol{\mu}_k),\Phi)} = \lambda\Phi(\mathbf{x}), \tag{R}$$

$$\boxed{\text{Therefore } \Phi(\mathbf{x}) = \sum_{k=1}^{N_{train}} \alpha_k u(\mathbf{x},\boldsymbol{\mu}_k).}$$

$$a_k = (u(\boldsymbol{\mu}_k),\Phi) = (u(\boldsymbol{\mu}_k), \textstyle\sum_{j=1}^{N_{train}} \alpha_j u(\boldsymbol{\mu}_j)) = \sum_{j=1}^{N_{train}} \alpha_j \underbrace{(u(\boldsymbol{\mu}_k), u(\boldsymbol{\mu}_j))}_{\mathbf{C}_{k,j}}$$

# Snapshot POD

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# Snapshot POD

$$(R) \Leftrightarrow \frac{1}{N_{train}} \sum_{k=1}^{N_{train}} u(\mathbf{x}, \boldsymbol{\mu}_k) \sum_{j=1}^{N_{train}} \alpha_j \mathbf{C}_{k,j} = \lambda \sum_{k=1}^{N} \alpha_k u(\mathbf{x}, \boldsymbol{\mu}_k)$$

gives for one $k = i$

$$\boxed{\frac{1}{N_{train}} \sum_{j=1}^{N_{train}} \mathbf{C}_{i,j} \alpha_j = \lambda \alpha_i.}$$

Thus, $\mathbf{C}\alpha = N_{train}\lambda\alpha = \lambda'\alpha$: the eigenvalues of the spatial covariance operator or the snapshot correlation matrix are the same (up to a factor $N_{train}$)!

# Discretization

$$\mathbf{C}\alpha = \lambda'\alpha \tag{C}$$

We impose $(\alpha_n)$ orthonormal.
Let $S = [u_1, \ldots, u_{N_{train}}] \in \mathbb{R}^{\mathcal{N} \times N_{train}}$. Then $\Phi = S\alpha$, and
$\|\Phi\|^2 = (S\alpha, S\alpha) = \alpha^T S^T S\alpha$
But $(C)$ implies that $\|\Phi\|^2 = \lambda'\alpha^T\alpha = \lambda'$

$$\text{Hence } \tilde{\Phi}(\mathbf{x}) = \frac{1}{\sqrt{\lambda'}} \sum_{k=1}^{N_{train}} \alpha_k u(\mathbf{x}, \boldsymbol{\mu}_k) \text{ ( after normalization)}$$

Let us denote in the following slides $\Phi = \tilde{\Phi}$ and $\lambda = \lambda'$.

# Discretization: Snapshot POD algorithm

1: Collect snapshots $u(\cdot, \mu_i), \ i = 1, \ldots, N_{train}$
2: Assemble snapshot matrix $S$
3: Compute correlation matrix $C = S^T S$ or $C = S^T M S$ ($M$= mass matrix)
4: Solve $C\alpha_i = \lambda_i \alpha_i, \ i = 1, \ldots, N_{train}$
5: Sort the eigenvalues
6: Retrieve first $N$ eigenvalues/eigenvectors
7: Build POD modes $\Phi_i = \frac{1}{\sqrt{\lambda_i}} S\alpha_i, \ i = 1, \ldots, N$

The space of the modes is contained into the snapshots' span: $N \leq N_{train}$
How do we choose $N$? $\lambda_1 >= \lambda_2 >= \ldots >= 0$

$$\boxed{\mathbb{E}[\|u - P_N u\|^2] = \sum_{k>N} \mathbb{E}[a_k^2] = \sum_{k>N} \lambda_k}$$

**Relativ Information Content (RIC) must be close to 0:**
$1 - \sum_{k=1}^{N} \lambda_k / \sum_{k=1}^{N_{train}} \lambda_k$

# TP VF5-TPFA

**Notations**

Mesh ($\mathcal{T}$) size: $h$

Sets of edges: $\mathcal{F}, \mathcal{F}_{ext}, \mathcal{F}_{int}, \mathcal{F}_K$

Normals: $\mathbf{n}_K, \mathbf{n}_{K\sigma}, \mathbf{n}_{KL}$

Volumes / Measures/Distances: $|K|, |\sigma|, d_{K\sigma}, d_{L\sigma}, d_{KL}$

**Finite Volume Methods**

Based on the conservation form of the PDE

Integrate the balance equation on each cell $\kappa$ and apply Stokes' formula:

$$\sum_{\text{edges of } \kappa} \text{outward flux} = \int_{\kappa} \text{source}.$$

Approximate each flux and write the discrete balance equation obtained.

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## Finite Volume Methods

Based on the conservation form of the PDE $\rightarrow$ Flux: total outward flux = the total internal source

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## Finite Volume Methods

Based on the conservation form of the PDE $\rightarrow$ Flux: total outward flux $=$ the total internal source

Integrate the equation on each cell $\kappa$ and apply Stokes' formula:

$$\int_K f(\mathbf{x})\ d(\mathbf{x}) = -\int_K \nabla \cdot \big(a(\boldsymbol{\mu})\nabla u\big) = \sum_{\sigma \in \mathcal{F}_K} \underbrace{-\int_\sigma a(\boldsymbol{\mu})\nabla u(\mathbf{x}) \cdot \mathbf{n}_{K,\sigma}\ d_\gamma(\mathbf{x})}_{\overline{F}_{K,\sigma}}$$

Approximate each flux and write the discrete balance equation obtained.

# TP VF5-TPFA

**Notations**

Mesh ($\mathcal{T}$) size: $h$

Sets of edges: $\mathcal{F}, \mathcal{F}_{ext}, \mathcal{F}_{int}, \mathcal{F}_K$

Normals: $\mathbf{n}_K, \mathbf{n}_{K\sigma}, \mathbf{n}_{KL}$

Volumes / Measures/Distances: $|K|, |\sigma|, d_{K\sigma}, d_{L\sigma}, d_{KL}$

Flux balance:

$$\sum_{\sigma \in \mathcal{F}_K} \overline{F}_{K,\sigma} = \int_K f(\mathbf{x})\ d(\mathbf{x}).$$

Flux conservativity:

$$\overline{F}_{K,\sigma} + \overline{F}_{L,\sigma} = 0 \text{ if } \sigma = K|L.$$

We want to find $u_h = (u_K)_{K \in \mathcal{T}} \in \mathbb{R}^{\mathcal{T}}$
Define $u_h \to F_{K,\sigma}(u_h)$ that approximates the flux and find $u_h \in \mathbb{R}^{\mathcal{T}}$ such that
$|K| f_K = \sum_{\sigma \in \mathcal{F}_K} F_{K,\sigma} \forall K \in \mathcal{T}$.

# TP VF5-TPFA

We want to find $u_h = (u_K)_{K\in\mathcal{T}} \in \mathbb{R}^{\mathcal{T}}$
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.

**Case of an interior edge** $\sigma \in \mathcal{E}_{int}, \ \sigma = K|L$

$$x_L - x_K = d_{KL}\,\mathbf{n}_{KL}.$$

If $x \in \sigma$,

$$(\nabla u(x)) \cdot \mathbf{n}_{KL} = \frac{u(x_L) - u(x_K)}{d_{KL}} + \mathcal{O}(h).$$

$$\Longrightarrow \overline{F}_{K,\sigma} = \underbrace{-\overline{A(\boldsymbol{\mu})}|\sigma|\frac{u(x_L) - u(x_K)}{d_{KL}}}_{F_{K,\sigma}(u_h)} + \mathcal{O}(h^2)$$

where $\overline{A}$ is the harmonic average: $\overline{A} = \frac{A(x_L)A(x_K)d_{KL}}{A(x_L)d_{K,\sigma}+A(x_K)d_{L,\sigma}}$

# TP VF5-TPFA

We want to find $u_h = (u_K)_{K\in\mathcal{T}} \in \mathbb{R}^{\mathcal{T}}$
Define $u_h \to F_{K,\sigma}(u_h)$ that approximates the flux and find $u \in \mathbb{R}^{\mathcal{T}}$ such that

$$|K| f_K = \sum_{\sigma\in\mathcal{F}_K} F_{K,\sigma}, \ \forall K \in \mathcal{T}$$

.

**Case of a boundary edge** $\sigma \in \mathcal{E}_{ext}$

$$x_\sigma - x_K = d_{K\sigma}\, \mathbf{n}_{K\sigma}.$$

$$(\nabla u(x)) \cdot \mathbf{n}_{K\sigma} \approx \frac{u(x_\sigma) - u(x_K)}{d_{K\sigma}} = \frac{0 - u(x_K)}{d_{K\sigma}} \qquad \text{(boundary condition)}$$

$$\Longrightarrow \overline{F}_{K,\sigma} = \underbrace{-|\sigma| A_K \frac{-u(x_K)}{d_{K\sigma}}}_{F_{K,\sigma}(u_h)} + \mathcal{O}(h^2)$$

# TP VF5-TPFA

Find $u_h = (u_K)_{K \in \mathcal{T}_h}$ such that for all $K$ in $\mathcal{T}_h$:

$$\sum_{\sigma \in F_K \cap F_{int}} \tau_\sigma (u_K - u_L) + \sum_{\sigma \in F_K \cap F_{ext}} \tau_\sigma u_K = \int_K f(x) dx$$

with Dirichlet boundary $u = 0$ on $\partial\Omega$, where $\tau_\sigma = |\sigma| \frac{A_K \ A_L}{A_L \ d_{K,\sigma} + A_K \ d_{L,\sigma}}$ on $F_{int}$ and $\tau_\sigma = |\sigma| \frac{A_K}{d_{K,\sigma}}$ on $F_{ext}$

We take here $\Omega = [0,1] \times [0,1]$ with a cartesian mesh.
$A(x, y; \mu) = 2\mu_1 + \mu_2 \ sin(x + y) \ cos(xy)$
$f(x, y; \mu) = \mu_3(1 - y) + \mu_4 x \ (1 - x)$

# TP VF5-TPFA

## POD-based Reduced Order Model with TPFA

- Complete the function `assemble_tpfa`. The TPFA solver must return the cell centers, the matrix $M$, and the vector $b$ such that $Mu = b$.
- Generate a training dataset:

  Use $N_{train} = 10$ snapshots and sample random parameters $\boldsymbol{\mu}$ with components in $[0, 1]$.
  Solve the full-order TPFA system for each sampled parameter.
  Store the resulting solutions as a snapshots list.
- Using the discrete $L^2$ inner product $(u, v)_{L^2} = \sum_K |K|\, u_K v_K$,

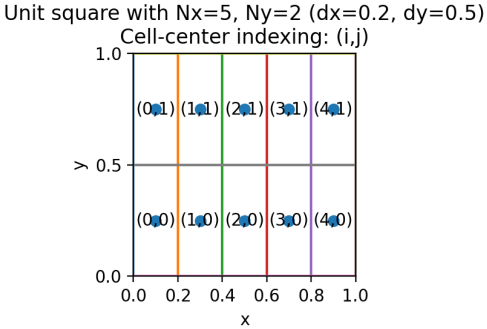
  Assemble the snapshot correlation matrix.
  Compute the reduced basis with a Proper Orthogonal Decomposition (POD).
  Verify that the reduced basis is orthonormal with respect to $(\cdot, \cdot)_{L^2}$.
- Determine how many modes $N$ are sufficient using the Relative Information Content.
- Write a function that computes the ROM projection coefficients of a given full-order solution $u$.
- Consider a new parameter $\boldsymbol{\mu}$. Write a function that computes the reduced-order approximation of the solution without computing the HF solution.
- Show that the obtained reduced system has the form $\widetilde{M}a = \widetilde{b}$, where $\widetilde{M}$ is of size $N \times N$ and $\widetilde{b}$ is of size $N$.

# TP VF5-TPFA

## POD-based Reduced Order Model with TPFA

- Test the reduced model for $\boldsymbol{\mu} = (0.6, 0.5, 0.2, 0.8)$.
- Compare the errors $\|u_{\mathrm{ref}} - u\|_{L^2}$ and $\|u_{\mathrm{ref}} - u_N\|_{L^2}$, where $u_{ref}$ is a refined solution.

# TP


Unit square with Nx=5, Ny=2 (dx=0.2, dy=0.5)
Cell-center indexing: (i,j)
(0,1)
(1,1)
(2,1)
(3,1)
(4,1)
(0,0)
(1,0)
(2,0)
(3,0)
(4,0)
1.0
0.5
0.0
y
0.0
0.2
0.4
0.6
0.8
1.0
x

# TP: FEM

with Kolmogorov $n$ width not small: $u(x, \mu) = tanh(\frac{x-\mu}{\delta})$.

# TP: FEM

FEM with Kolmogorov $n$ width not small: Burgers equation:
Viscous Burgers (1D) with periodic BC using scikit-fem

$$u_t + nu * u * u_x - eps * u_x x = 0 in(0, T]x[-1, 1]$$
$$u(0, x) = u0(x) = lam + sin(x)$$
$$uperiodicatx = -1andx = 1$$

Time stepping: IMEX (explicit convection, implicit diffusion)
$(M + dt * eps * K)u^{n+1} = Mu^n - dt * F(u^n)$ where
$F_i(u^n) = nu * v_i * (u^n) * (u^n)_x dx$